

Reference Guide

1. Spring Boot (MVC Architecture)

What is Spring Boot?

Spring Boot is a Java framework used to build production-ready web applications quickly. It follows the MVC (Model-View-Controller) architectural pattern.

Key Concepts:

- MVC Architecture
- REST Controllers
- Dependency Injection
- Auto Configuration
- Embedded Servers (Tomcat)

Official Documentation:

- <https://spring.io/projects/spring-boot>
- <https://docs.spring.io/spring-boot/docs/current/reference/html/>

MVC Architecture Reference:

- <https://www.baeldung.com/spring-mvc-tutorial>
- <https://www.geeksforgeeks.org/spring-mvc-framework/>

2. Hibernate (JPA)

What is Hibernate?

Hibernate is an ORM (Object Relational Mapping) framework that implements JPA (Java Persistence API). It maps Java objects to database tables.

Key Concepts:

- Entity Classes
- @Entity, @Id, @GeneratedValue
- One-to-One, One-to-Many Relationships
- JPQL
- Fetch Types (EAGER / LAZY)

Official Documentation:

- <https://hibernate.org/orm/>
- https://docs.jboss.org/hibernate/orm/current/userguide/html_single/

JPA Specification:

- <https://jakarta.ee/specifications/persistence/>

Tutorials:

- <https://www.baeldung.com/hibernate-tutorial>
- <https://www.geeksforgeeks.org/hibernate-tutorial/>

3. Relational Databases (MySQL)

MySQL

MySQL is a widely used open-source relational database management system.

Official Website:

- <https://www.mysql.com/>

Documentation:

- <https://dev.mysql.com/doc/>